SECR3253-02 NETWORK PROGRAMMING

Network Automation Project

Personal Reflection

Group: Nahj

GROUP MEMBERS	MATRIC NUMBER
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1. My Contribution to the Project

My role in Group Nahj was to handle the network routing configuration and device information retrieval, which formed a core part of the network automation requirements in the assignment. Specifically, I was responsible for writing and testing `playbooks/net_routing.yml`, working on the `feature/net-routing` branch.

My playbook covered three main tasks. The first was configuring interface descriptions on both simulated LAN interfaces — Loopback0 representing LAN1 and Loopback1 representing LAN2. Adding meaningful descriptions to interfaces is an important network management practice as it makes the device configuration readable and helps identify the purpose of each interface at a glance. The second task was adding a static route to route traffic from the LAN1 subnet (192.168.10.0/24) to the LAN2 subnet (192.168.20.0/24) through the router. This simulated a real routing scenario between two separate LAN segments. The third task was retrieving and displaying device information from the router using the `ios_command` module, collecting the output of `show running-config`, `show version`, and `show ip interface brief` to verify that all configurations were applied correctly.

Beyond the technical work, I participated in the group's Git workflow by working on my own branch, making incremental commits across multiple days, and opening a Pull Request for the group leader to review before merging into main.

2. Challenges Encountered During Collaboration

The biggest challenge I faced personally was getting familiar with Ansible and how it communicates with network devices. Before this project, I had written basic scripts but never used a configuration management tool like Ansible. Understanding the structure of a playbook hosts, gather_facts, vars_files, tasks — took some time to get right, especially since network device playbooks behave differently from regular Linux host playbooks. For example, I learned that `gather_facts: no` is required for network devices because Ansible cannot gather facts the same way it does from a Linux system.

Another challenge was understanding the `ios_config` module and how it handles idempotency. Unlike raw SSH commands, `ios_config` checks the existing configuration before applying changes, which means the syntax of the commands passed must closely match how they appear in the running configuration. I initially had some mismatches that caused unexpected warnings during playbook runs.

From a collaboration standpoint, coordinating with the rest of the group through Git branches and Pull Requests was a new experience. There were a few times where I was unsure whether my changes would conflict with someone else's work, but the clear file ownership structure that Omar set up at the start meant that each member had their own isolated file to work on, which avoided most of those issues.

3. What I Learned from the Project

This project was my first real exposure to network automation and it significantly changed how I think about network configuration. Before this, I associated network configuration with logging into devices one by one through a terminal and typing commands manually. Seeing Ansible push configuration to a Cisco IOS XE router automatically and having it be repeatable and version-controlled was genuinely eye-opening.

I also learned the value of using Git properly in a team setting. Having separate branches for each member, committing changes incrementally with meaningful messages, and going through a Pull Request review process before merging felt like overhead at first, but it became clear why these practices exist when working in a group of five people. It kept everyone's work isolated and made it easy to track who contributed what and when.

Working with the `cisco.ios` Ansible collection specifically taught me that network automation is not just about knowing CLI commands it is about understanding how those commands map to structured automation modules and how to express network intent in YAML rather than imperative terminal sessions. This is a skill I intend to build on further, as network automation is increasingly relevant in modern infrastructure roles.